Galois Groups of Trinomials with Small Coefficients

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2026

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- Bary-Soroker, Shusterman & Zannier (2018).

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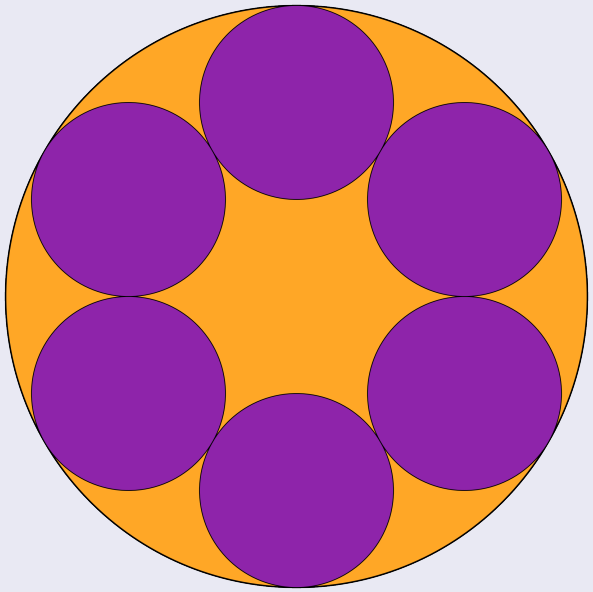
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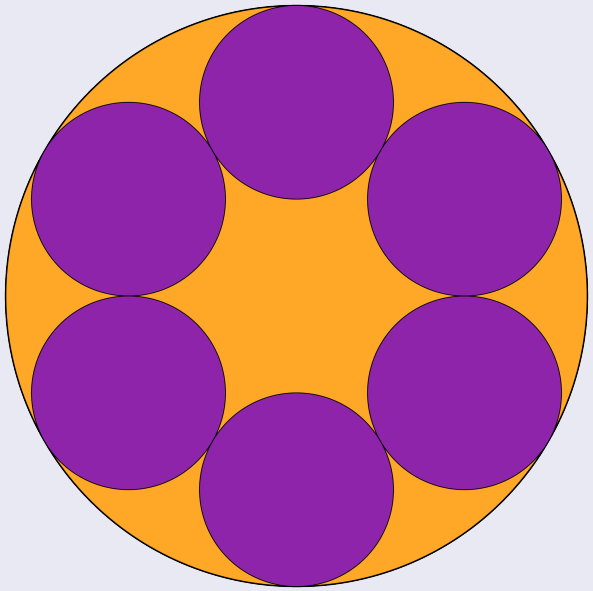
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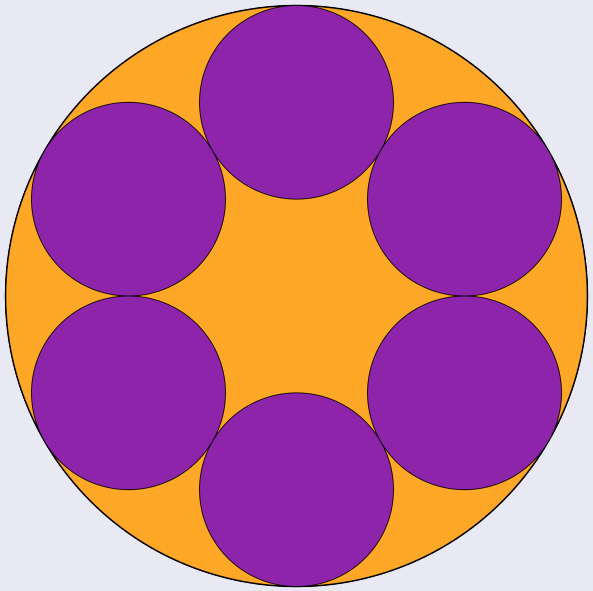
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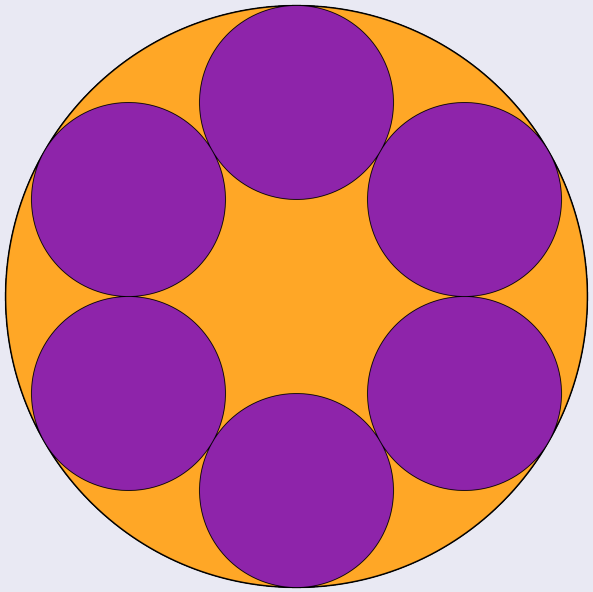
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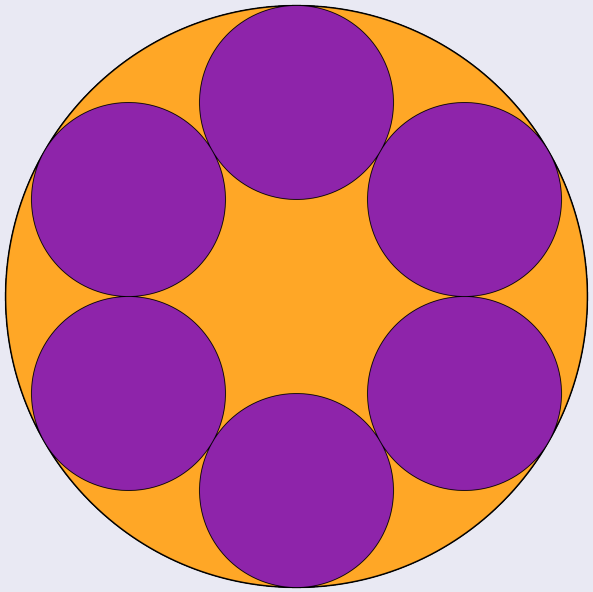
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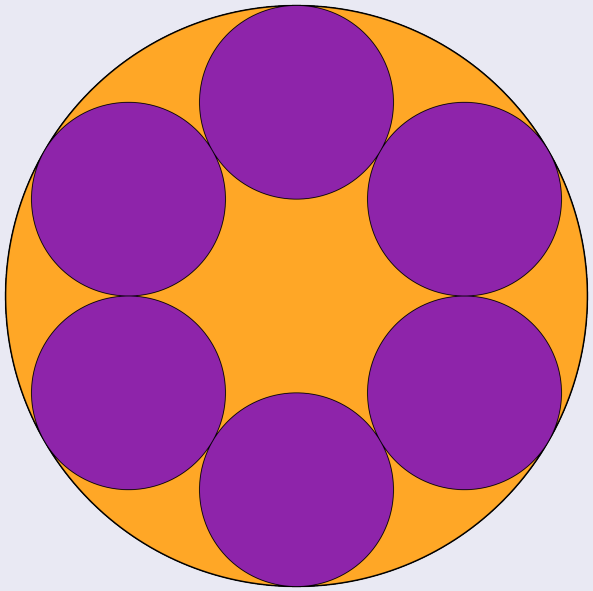
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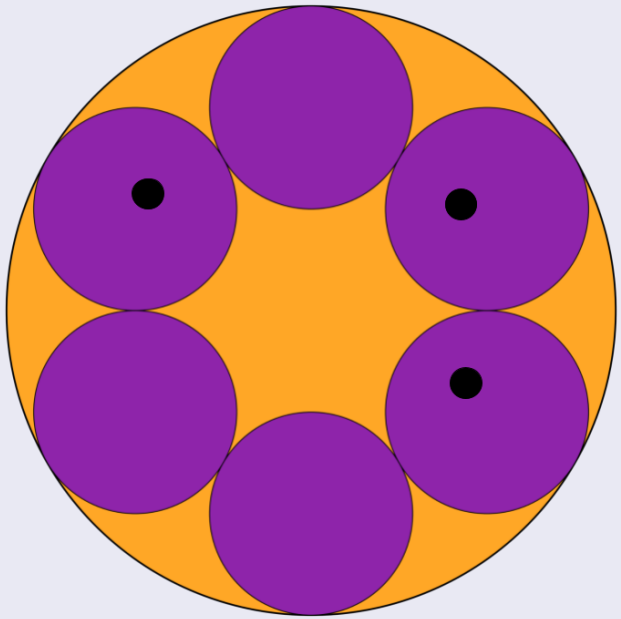
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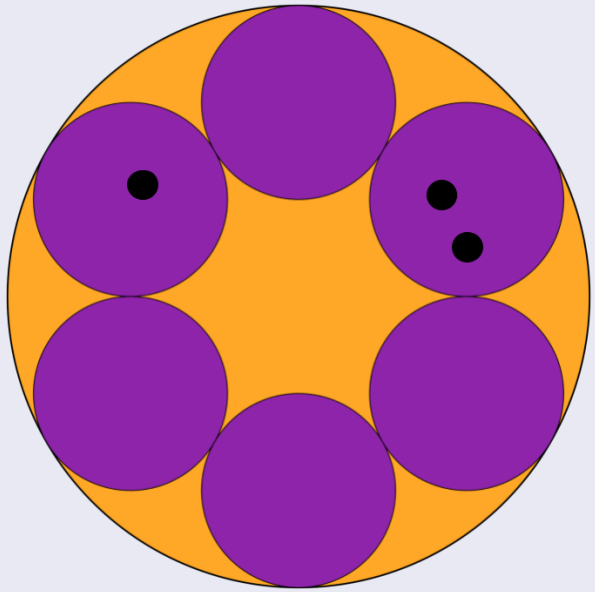
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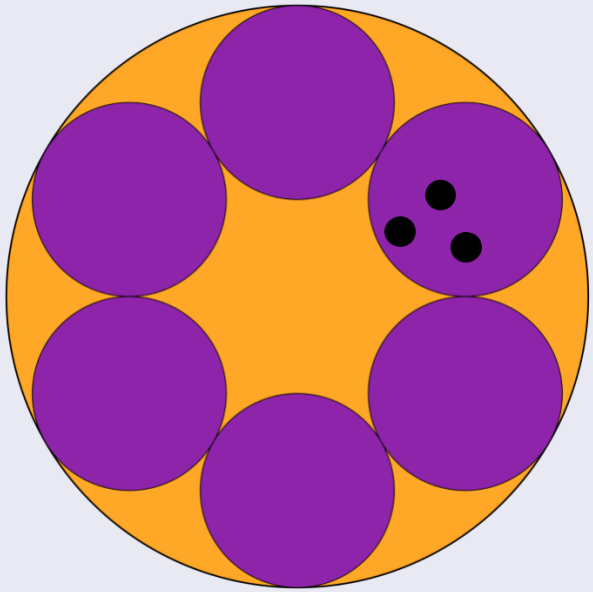
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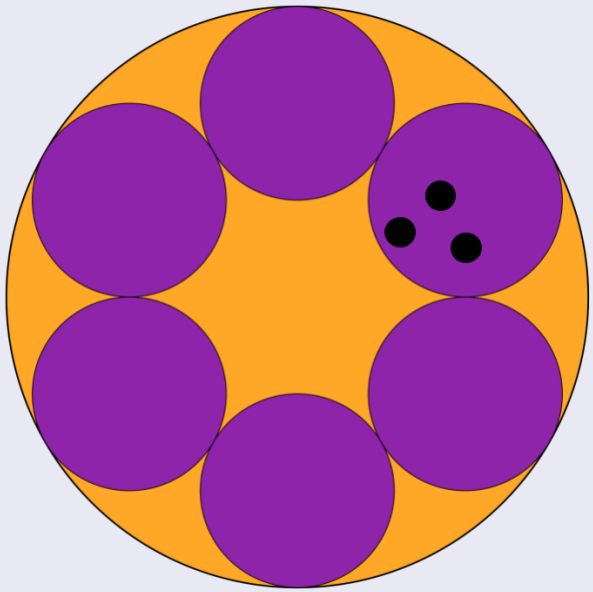


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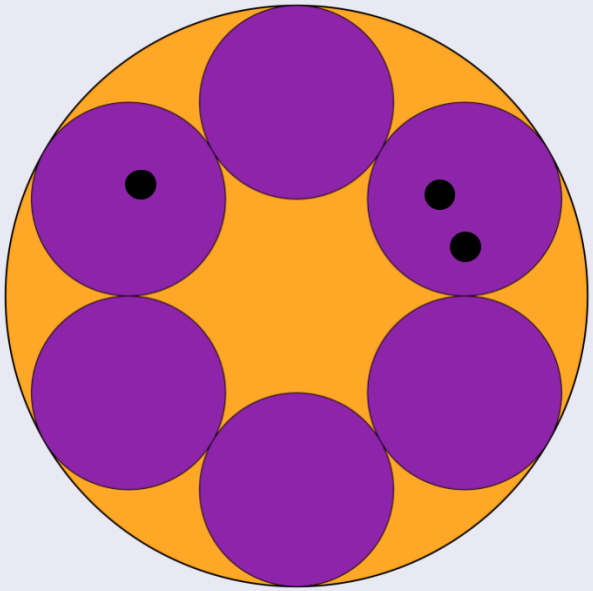
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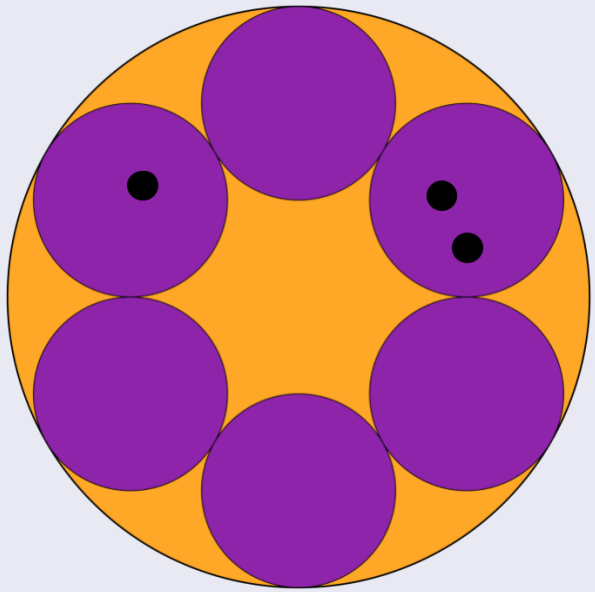
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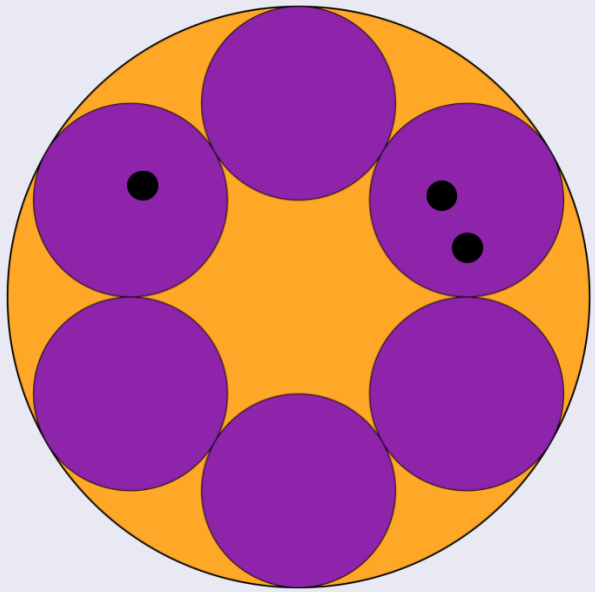
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Permutation Groups Theory Fact:

A transitive permutation group which is generated by transpositions is the full symmetric group.

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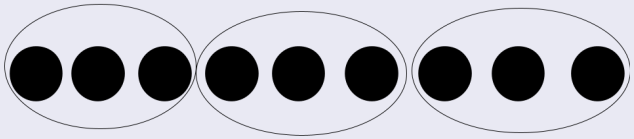
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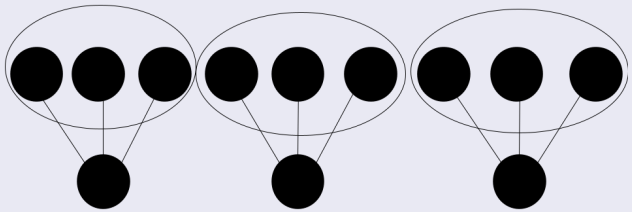
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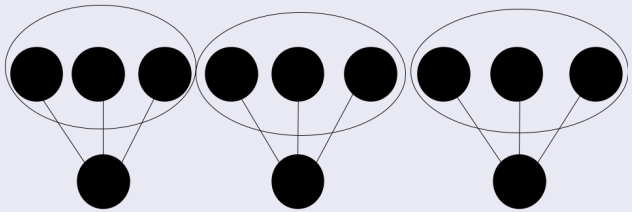
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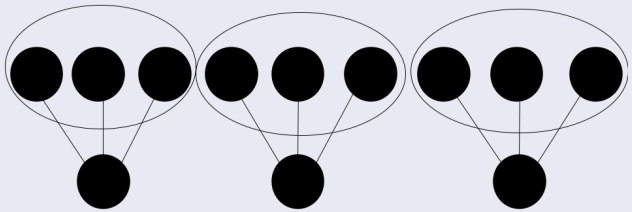


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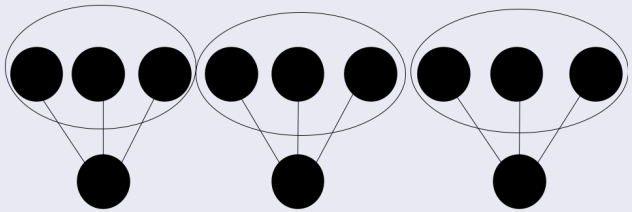


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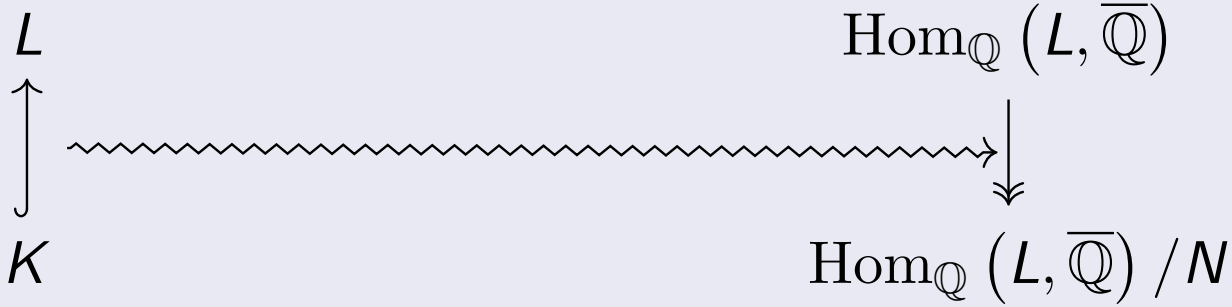
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Odlyzko (1977)

$$\mathrm{rd}\,(K) > 22.381 - O\left(n^{-2/3}\right)$$

Future Results

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Conjecture / Pre-Theorem

For any integers $0 < k < n$, such that $\gcd(n, k) = 1$, and $n > 7$, let

$$f(x) = x^n \pm 2x^k \pm 1$$

Then one of two cases occures:

- $f(1) f(-1) = 0$ and the Galois group of f is S_{n-1}.
- $f(1) f(-1) \neq 0$ and the Galois group of f is S_n.